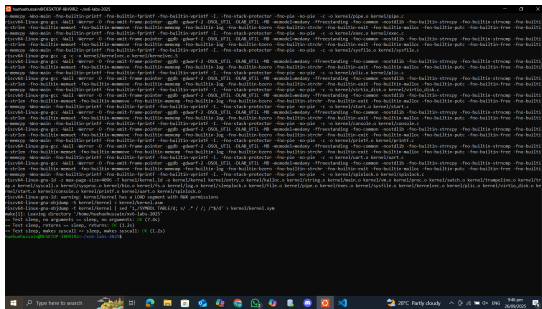


Problem Set 1

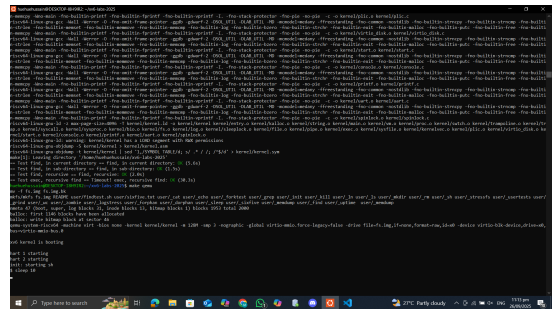
Muhammad Hussain - 29004
Operating Systems

September 26, 2025

Sleep

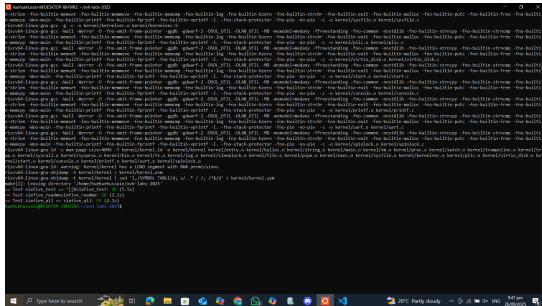


(a) Make Grade(sleep)

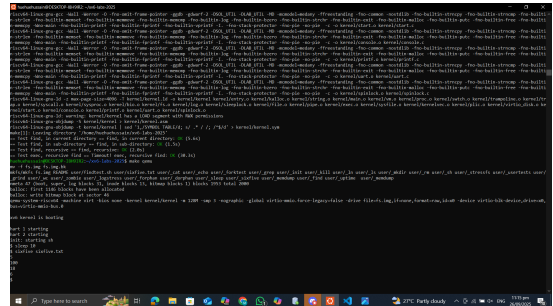


(b) Output(sleep)

Sixfive

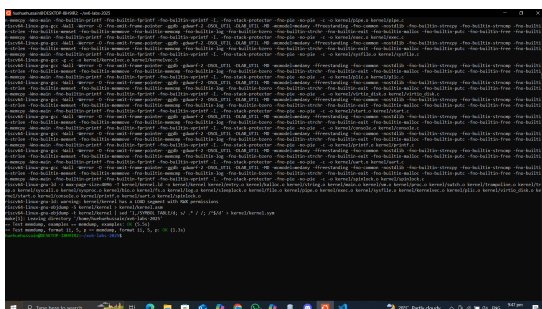


(a) Make Grade(sixfive)

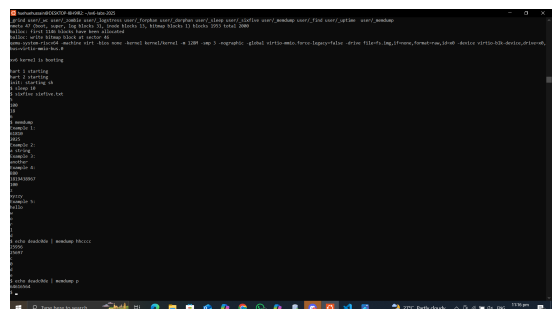


(b) Output(sixfive)

Memdump



(a) Make Grade(memdump)



(b) Output(memdump)

Find

```

root@kali:~/Documents# cat test.py
#!/usr/bin/perl
use strict;
use warnings;

my $url = "http://10.10.10.10:8080/";
my $path = "/";
my $method = "GET";

my $headers = {
    "Host" => "10.10.10.10:8080",
    "User-Agent" => "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36",
    "Accept" => "text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8",
    "Accept-Language" => "en-US,en;q=0.9",
    "Accept-Encoding" => "gzip, deflate, br",
    "Connection" => "close",
    "Upgrade-Insecure-Requests" => 1
};

my $response = http_get($url.$path, $headers, $method);

if ($response =~ /HTTP\/1.1 200 OK/) {
    print "Success: HTTP 200 OK\n";
} else {
    print "Error: HTTP status not 200 OK\n";
}

sub http_get {
    my ($url, $headers, $method) = @_;
    my $socket = socket($url);
    my $request = "GET $path HTTP/1.1\n";
    foreach my $header (keys %$headers) {
        $request .= "$header: $headers{$header}\n";
    }
    $request .= "\n";
    send($socket, $request, $method);
    my $data = receive($socket);
    close($socket);
    return $data;
}

sub socket {
    my ($url) = @_;
    my ($host, $port) = ($url =~ /^http:\/\/(.+):(\d+)/);
    my $socket = socket(AF_INET, SOCK_STREAM, 0);
    setsockopt($socket, SOL_SOCKET, SO_REUSEADDR, 1);
    connect($socket, sockaddr_in($port, $host));
    return $socket;
}

sub send {
    my ($socket, $request, $method) = @_;
    my $data = $request.$method;
    send($socket, $data);
}

sub receive {
    my ($socket) = @_;
    my $data = "";
    while (1) {
        my $chunk = recv($socket, my $buffer, 1024, 0);
        if ($chunk == 0) {
            last;
        }
        $data .= $buffer;
    }
    return $data;
}

```

(a) Make Grade(find)

[illegible]

(b) Output(find)

Exec

[illegible]

(a) Output(exec)